Implementation Of SHANNON FANO ELIAS ENCODING ALGORITHM Using LabVIEW

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Shannon Fano Elias encoding algorithm is a precursor to arithmetic coding in which probabilities are used to determine code words. It is a lossless coding scheme used in digital communication. Probability theory has played an important role in electronics communication systems. It is because information in a signal is usually accompanied by noise. At the receiver end, when we talk about recovering the message signal, we are basically talking about estimating the original signal from the received signal accompanied by noise. In digital communication, we talk about decoding the receiving bits of a message with some probability of errors.

This article describes a simple construction procedure that uses cumulative distribution function (CDF) to generate code words using LabVIEW. The code word length of Shannon Fano Elias coding satisfies Kraft's inequality. Therefore it is used to construct decodable code for the information source.

Shannon Fano Elias code procedure can be applied to a sequence of random variables. Probability and CDF can be calculated sequentially with symbols in a block or group of data, ensuring that the calculation grows linearly with block length. Direct application of Shannon Fano Elias code also needs arithmetic whose precision grows with block size.

Encoding algorithm

In digital communication, the source encoder converts input, that is, symbol sequence, into a binary sequence of 0s and 1s. Each source symbol is represented by some sequence of coded symbols called code words.

Shannon Fano Elias encoding algorithm procedure is given as follows:

Step 1. Arrange the symbols according to decreasing probabilities.

Step 2. Calculate cumulative probability.

$\sigma1 = 0$; $\sigma2 = p1 + \sigma1$; $\sigma3 = p2 + \sigma2$; etc...

Step 3. Calculate modified cumulative distribution function.

$Fbar(x) = \sigma i + p(xi)/2$

Step 4. Find the length of the code word.

$l(x) = [\log 1/p(x)] + 1$

Step 5. Generate the code word by finding the binary of Fbar(x) with respect to length l(x).

Example. Consider a source with five symbols with probabilities 0.3, 0.25, 0.2, 0.15 and 0.1 (see table).

EXAMPLE

Source symbols	Pi	σi	Fbar(x)	l(x)	Code word
x1	0.3	0	0.15	3	001
x2	0.25	0.3	0.425	3	011
x3	0.2	0.55	0.65	4	1010
x4	0.15	0.75	0.825	4	1101
x5	0.1	0.9	0.95	5	11110

Implementation of algorithm in LabVIEW

LabVIEW programs or virtual instruments (VIs) have front panels and block diagrams. Front panel (Fig. 1) is the user interface. Block diagram (Fig. 2) is the programming behind the user interface. After building the front panel, the code is added in the

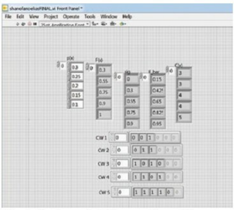

Fig. 1: Front panel of Shannon Fano Elias encoding

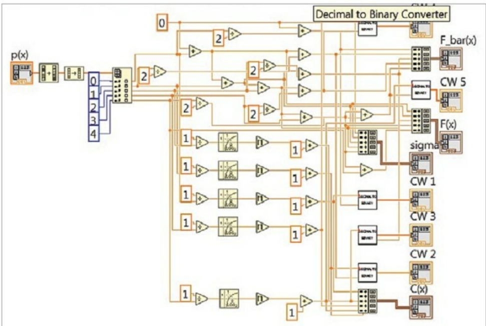

Fig. 2: Block diagram of Shannon Fano Elias encoding